

Bluetooth Low Energy (BLE)

Objective:

To understand the Bluetooth Low Energy (BLE) functionality of the Arduino Nano 33 BLE Sense Lite and establish wireless communication between the board and a mobile device.

(Note:

1. install ArduinoBLE Library
2. nRF Connect Application in Android Smartphone)

Procedure:

1. Installed the ArduinoBLE library using the Library Manager.
2. Opened the BLE LED Peripheral example.
3. File → Examples → ArduinoBLE → Peripheral → LED
4. Uploaded the program to the Arduino Nano 33 BLE Sense Lite.
5. Installed the nRF Connect application on a smartphone.
6. Enabled Bluetooth on the smartphone.
7. Opened nRF Connect and scanned for nearby BLE devices.
8. Observed the Arduino board advertising itself as "LED"
9. Send commands through the application and observe the response.
01 → Turn LED ON
00 → Turn LED OFF

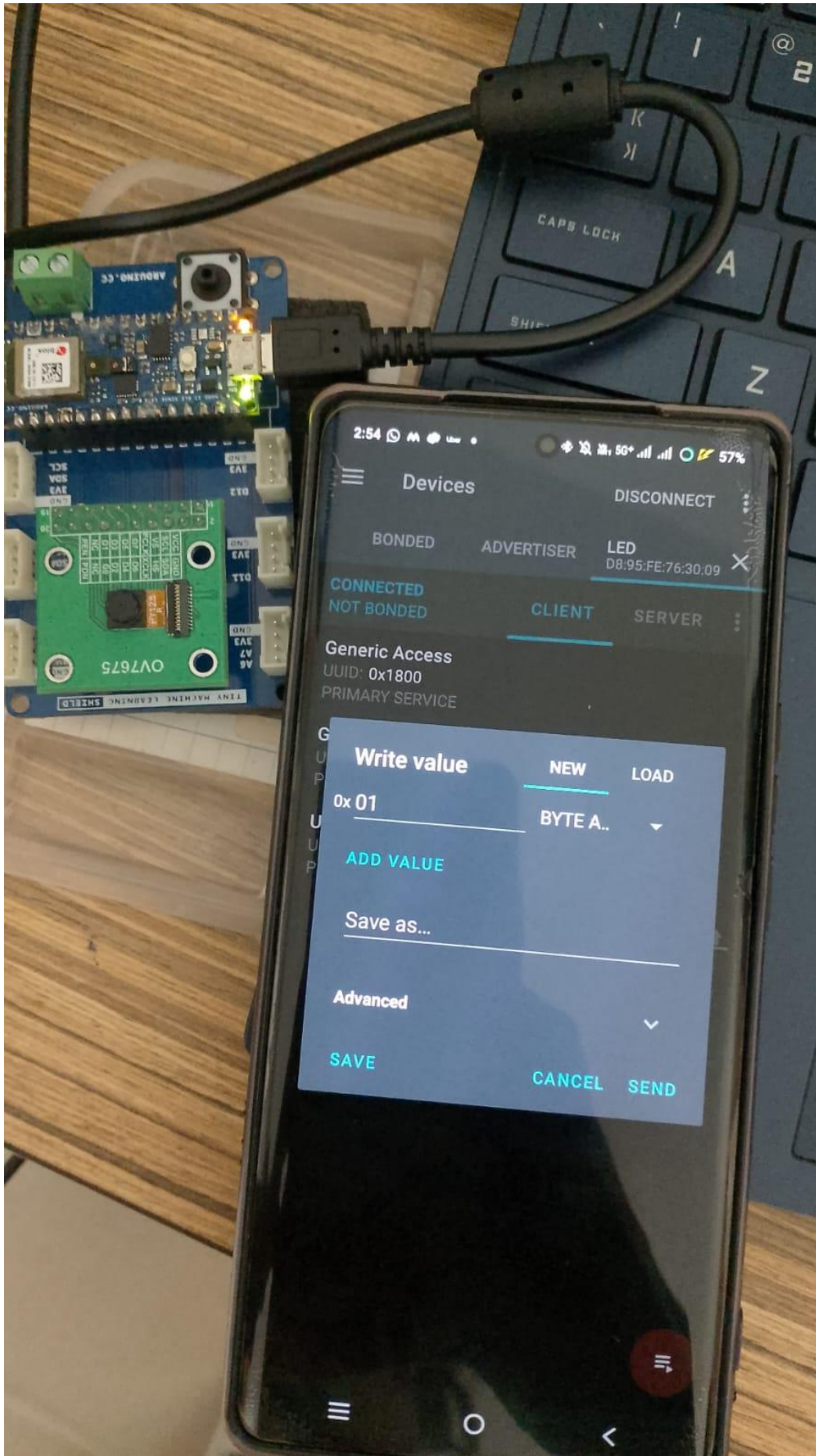
Working Principle:

The Arduino Nano 33 BLE advertises itself as a Bluetooth device. A smartphone can discover and connect to it. Once connected, data can be exchanged wirelessly between the Arduino and the mobile application.

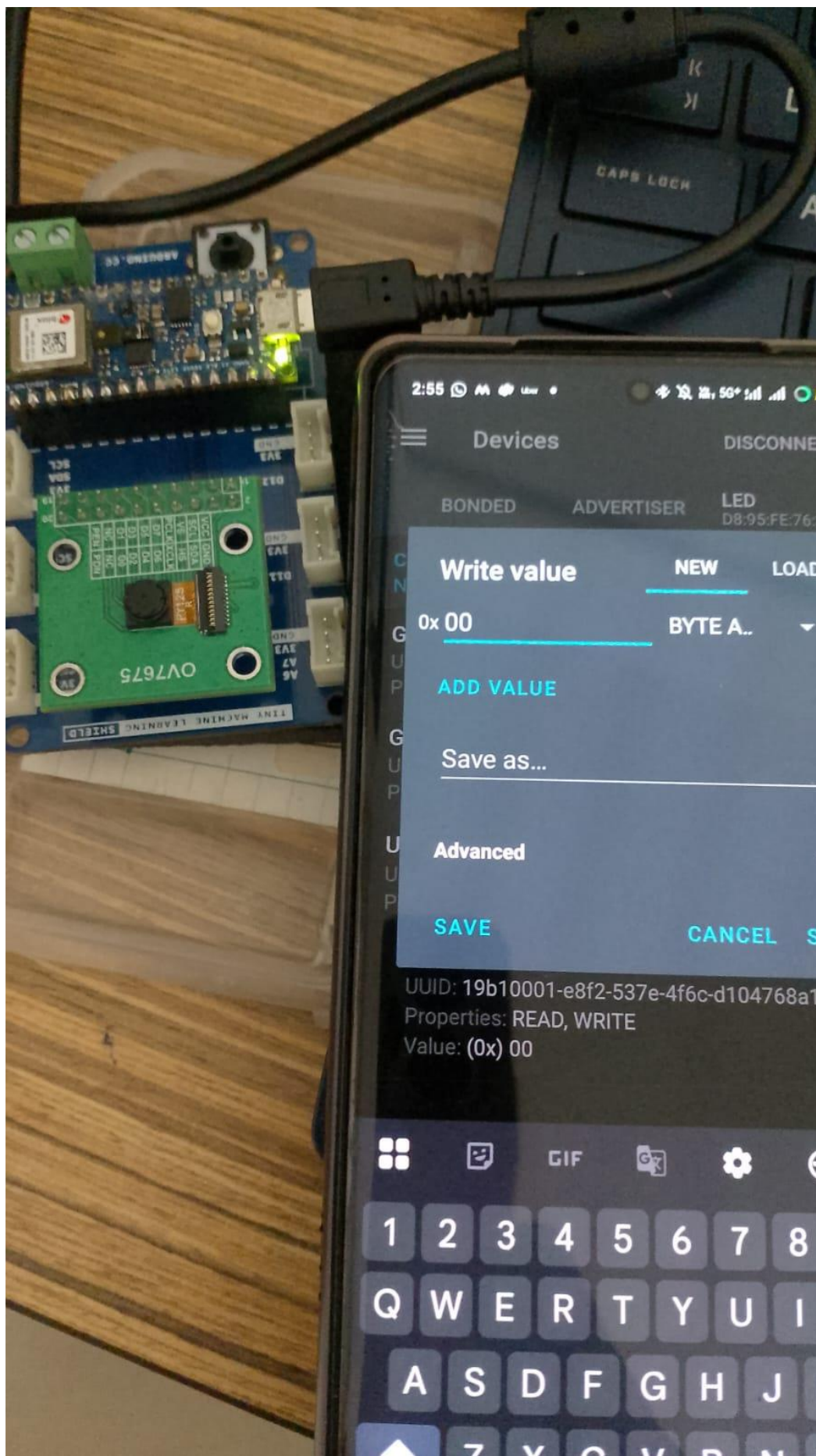
Observation:

The Arduino board was detected by the smartphone. Bluetooth connection was successfully established. Data was transmitted wirelessly between the Arduino and the mobile application.

When we send 01:



When we send 00:



(Note: In the "**0x New value**" box, type: 00 then tap **SEND**. Then orange/yellow LED turn OFF)

